

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2025-P1-Model)** **Date: 15-04-2026**

Time: 09.00Am to 12.00Pm **GTA-24** **Max. Marks: 180**

15-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2025-P1)_GTA-24_Syllabus,

PHYSICS : TOTAL SECOND YEAR SYLLABUS

CHEMISTRY : TOTAL SECOND YEAR SYLLABUS

MATHEMATICS : TOTAL SECOND YEAR SYLLABUS

Rankers Academy JEE
Sri Chaitanya Educational Institutions INDIA

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2025-P1-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with NUMERICAL VALUE Type	+4	0	6	24
Sec – IV(Q.N : 14 – 16)	Matching Type	+4	-1	3	12
Total				**16**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 17 – 20)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 21 – 23)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 24 – 29)	Questions with NUMERICAL VALUE Type	+4	0	6	24
Sec – IV(Q.N : 30 – 32)	Matching Type	+4	-1	3	12
Total				**16**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 37 – 39)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 40 – 45)	Questions with NUMERICAL VALUE Type	+4	0	6	24
Sec – IV(Q.N : 46 – 48)	Matching Type	+4	-1	3	12
Total				**16**	**60**

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MATHEMATICS

Max Marks: 60

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

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1. Let $P(3\sec\theta, 2\tan\theta)$ and $Q(3\sec\phi, 2\tan\phi)$ where $\theta+\phi=\frac{\pi}{2}$, be two distinct points on the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$. Then the absolute value of ordinate of the point of intersection of the normals at P and Q is

A) $\frac{11}{3}$ **B)** $\frac{14}{3}$ **C)** $\frac{13}{2}$ **D)** $\frac{17}{2}$

2. If $\int \frac{x^3+x^2+x}{\sqrt{15+12x+10x^2}}dx=\frac{g(x)}{k}\sqrt{15+12x+10x^2}+C$, where C is the constant of integration and $g(1)=1$ then $(g'(1)+g''(1)+k)$ is equal to

A) 30 **B)** 32 **C)** 34 **D)** 36

3. The value of $\int_0^{\pi/2} \sin x \ln(\sin x)dx$ is

A) $\ln\left(\frac{2}{e}\right)$ **B)** $\ln\left(\frac{e}{4}\right)$ **C)** $-\pi\ln 2$ **D)** $-\frac{\pi}{2}\ln 2$

4. If two circles, each of radius 5 units, touch each other at $(1,2)$ and the equation of their common tangent is $4x+3y=10$, then equation of one of these two circles, which contain origin inside can be

A) $x^2+y^2-10x-10y+25=0$ **B)** $x^2+y^2+6x+2y-15=0$

C) $x^2+y^2+2x+6y-15=0$ **D)** $x^2+y^2+10x+10y+25=0$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

5. Consider a cube of side length 9 units. Let (x, y, z) be coordinates of points on or inside the cube such that $x, y, z \in I$ and $0 \le x, y, z \le 9$. If total number of ways of selecting two distinct points among these such that their mid-point is also having integral coordinates is N, then which of the following option(s) is/are correct?

A) N is divisible by 30 **B)** N is divisible by 31

C) N is divisible by 32 **D)** Number of positive factors of N is 40

6. Let $y = f(x)$, $y = g(x)$ be two positive real valued differentiable functions in R such that $g(x) = \int_{-\infty}^{x} f(t)dt$, $f(0) = 1, g(0) = 4$ and tangents to $y = f(x)$ and $y = g(x)$ at their points with equal abscissa always intersect on x axis, then which of the following option(s) is/are correct?

A) $\int_{-1}^{1} \frac{x^2+4}{4+g(x)}dx = \frac{13}{3}$ **B)** $\int_{-1}^{1} \frac{x^2+4}{4+g(x)}dx = \frac{13}{12}$

C) $f(x) - g(x)$ is decreasing in $[-1,1]$ **D)** $f(x) - g(x)$ is increasing in $[-1,1]$

7. Let $z_1, \bar{z}_1, z_2, \bar{z}_2, z_3, \bar{z}_3, z_4, \bar{z}_4, z_5, \bar{z}_5$ be the roots of the equation $x^{10} + (13x-1)^{10} = 0$.

If $N = \sum_{i=1}^{5} \frac{1}{|z_i|^2}$, then which of the following option(s) is/are correct?

A) the digit in the units place of N is 0 **B)** the sum of the digits in N is 13

C) the digit in the 10's place of N is 5 **D)** the product of digits of N is 0

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SECTION-III
(NUMERICAL VALUE)

This section contains **SIX (06)** questions.

- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

8. Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the parabola $y^2=2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangent to the ellipse at P and Q intersect the x-axis at R and tangent to parabola at P and Q intersect the x-axis at S. If the area of quadrilateral PRQS is A then determine the value of $\left(\frac{2}{5\sqrt{3}}A\right)$

9. Consider a regular heptagon $A_1A_2A_3A_4A_5A_6A_7$ inscribed in a unit circle.

Let $p_1=(A_1A_2)(A_1A_3)(A_1A_4)(A_1A_5)(A_1A_6)(A_1A_7)$

$p_2=(A_2A_3)(A_2A_4)(A_2A_5)(A_2A_6)(A_2A_7)$

$p_3=(A_3A_4)(A_3A_5)(A_3A_6)(A_3A_7)$

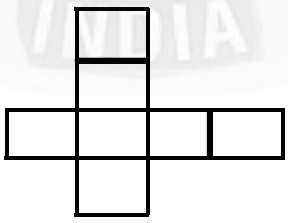
$p_4=(A_4A_5)(A_4A_6)(A_4A_7)$

$p_5=(A_5A_6)(A_5A_7)$ and $p_6=A_6A_7$ where A_iA_j is length of the segment joining A_i and A_j then value of product $(p_1p_2p_3p_4p_5p_6)$ is $(7)^k$, then the value of k is

10. The value of the limit 'L' is $e^{\frac{-1}{K}}$, then the value of K if

$$L=\lim_{n\to\infty} n^{\frac{-1}{2}\left(1+\frac{1}{n}\right)}\left(1^1.2^2.3^3....n^n\right)^{\frac{1}{n^2}}$$

11. The numbers $2,3,4,5,6,7,8$ are to be placed, one per square, in the diagram shown so that the sum of the four numbers in the horizontal row equals 21 and the sum of the four numbers in the vertical column also equals 21. In how many different ways can this be done?



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12. If a, b and c are three numbers selected at random simultaneously from the set {1, 2, 3, …, 2019}, then probability that $\text{Amp}\left(\frac{a-b}{a-c}\right)=0$ is $\frac{2}{k}$, then the value of k is

(where Amp(z) denotes principal argument of complex number z)

13. If the coefficient of x^2 in the expansion of the product $(1-x)(1+2x)(1-3x).....(1+14x)(1-15x)$ is C then the value of C is,

SECTION – IV
(MATCHING TYPE)

This section contains **THERE (03)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

14. A person buys 50 distinct lotteries, in each of which his chance of winning a prize is $\frac{1}{100}$.

The probabilities of winning a prize in different cases are given below, match them to their respective probabilities.

Column I		Column II	
A)	At least once	**P)**	$\frac{1}{2}\left(\frac{99}{100}\right)^{49}$
B)	Exactly once	**Q)**	$\frac{15976(99)^{48}}{100^{50}}$
C)	At least twice	**R)**	$1-\left(\frac{99}{100}\right)^{50}$
D)	At most twice	**S)**	$1-\frac{149}{100}\left(\frac{99}{100}\right)^{49}$

A) A-P, B-Q, C-S, D-Q **B)** A-R, B-R, C-Q, D-S

C) A-R, B-P, C-Q, D-S **D)** A-R, B-P, C-S, D-Q

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15. Consider a parabola $y^2 = lx(l > 0)$ with its vertex at the origin 'O'.

Column – I		Column – II	
A)	If origin is joined to any point P on the given parabola and a line perpendicular to OP through P meet the axis at Q, then projection of PQ on the axis is equal to	P)	$\frac{l}{4}$
B)	A normal is drawn at any arbitrary point P on the parabola which meets the axis at G. Locus of the middle point of PG is a parabola whose latus rectum is	Q)	$\frac{l}{2}$
C)	A variable parabola of latus rectum l, touches the given parabola, the axes of the two curves being parallel. The locus of the vertex of the variable parabola is another parabola is another parabola whose latus rectum is	R)	l
D)	The locus of the mid point of the focal radii of a variable point P moving on the given parabola, is a parabola whose latus rectum is	S)	$2l$

A) $A-P, B-R, C-S, D-Q$ **B)** $A-R, B-P, C-Q, D-S$

C) $A-R, B-P, C-S, D-Q$ **D)** $A-R, B-Q, C-S, D-P$

16.

Column – I		Column – II	
A)	Number of real solution(s) of the equation $\sum_{n=0}^{\infty}\left(\sin^2 x\right)^n + \sum_{n=0}^{\infty}\left(\cos^2 x\right)^n = 4$ in $(0, 2\pi)$	P)	1
B)	If the range of m for which the equation $\operatorname{cosec}^{-1} x = mx$ has exactly two solution is $\left(0, \frac{\lambda\pi}{10}\right]$, then λ is equal to	Q)	2
C)	Sum of integral solutions of the equality $[x]^2 = [x+6]$ is, (where [.] represent greatest integer function)	R)	4
D)	The number of real solution(s) of the equation $\left\lvert\operatorname{cosec}\left(\sin\left(\sin^{-1} x\right)\right)\right\rvert = \left(\frac{d}{dx}\left(\frac{x^{10}}{10} + \frac{x^8}{8} + x^2 - x^9 - \frac{9x^7}{7}\right)\right)_{x=9}$ is/are	S)	5

A) $A \to R; B \to S; C \to P; D \to Q$ **B)** $A \to S; B \to R; C \to P; D \to Q$

C) $A \to R; B \to P; C \to S; D \to Q$ **D)** $A \to R; B \to S; C \to Q; D \to P$

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PHYSICS **Max Marks: 60**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

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Negative Marks : −1 In all other cases

17. A capacitor is made of two infinite coaxial metal cylinders with a dielectric material in the space between the cylinders have radii a and b, respectively. The material has dielectric constant which varies with radius r as $k = \eta r$ where η is a constant. Find the capacitance per unit length of the system.

A) $\dfrac{2\pi \in_0 \eta}{\ell n\left(\dfrac{b}{a}\right)}$ **B)** $\dfrac{2\pi \in_0}{3\eta}\left(b^3 - a^3\right)$

C) $2\pi \in_0 \eta \dfrac{(ab)}{(b-a)}$ **D)** $\pi \in_0 \eta\left[b^2 - a^2\right]$

18. Consider the following three statements

Statement 1: In photoelectric effect, even for monochromatic incident radiations, the photoelectrons are emitted with a spread of velocities. (for frequency greater than cut off frequency)

Statement 2: Photoelectrons are emitted without delay once the incident light reaches the surface of the emitter. (for frequency greater than cut off frequency)

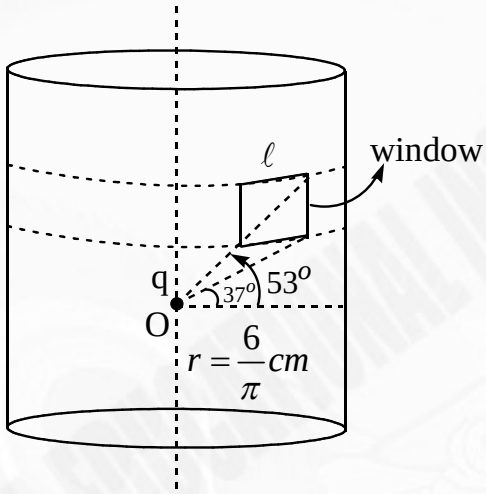
Statement 3: Frequency of the monochromatic light (well above the cut off frequency), that is incident on the emitter in a photoelectric setup is increased while keeping the intensity constant. It results in decrease in magnitude of stopping potential.

Regarding the statements, select the correct order of true and false.

A) TTF **B)** TFT **C)** FTT **D)** TFF

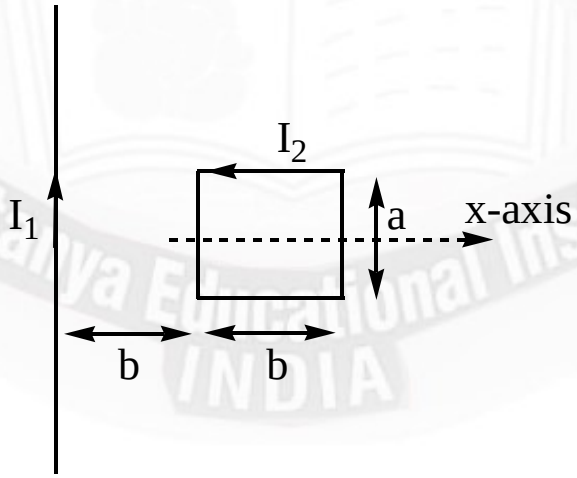
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19. In the given figure, a window is cut from a cylindrical shell. A point charge q is placed at point O on the axis as shown. If electric flux from the window is $\frac{q}{60\varepsilon_0}$, then the length ℓ of the window will be. Radius of cylinder, $r = \frac{6}{\pi} cm$.



A) 1 cm **B)** 2 cm **C)** π cm **D)** 2π cm

20. A rectangular wire loop of side lengths a and b is placed at a distance b in front of a long straight current carrying wire as shown. Initially the wire and the loop are coplanar. Now the loop is turned by 90° slowly about x-axis shown by a mechanical agent. The work done by the agent on the loop provided the currents I_1 and I_2 are maintained constant in process is



A) $-\frac{\mu_0 I_1 I_2 a}{2\pi}\ln 2$ **B)** $\frac{\mu_0 I_1 I_2 a}{2\pi}\ln 2$ **C)** $\frac{\mu_0 I_1 I_2 a}{4\pi}\ln 2$ **D)** $-\frac{\mu_0 I_1 I_2 a}{4\pi}\ln 2$

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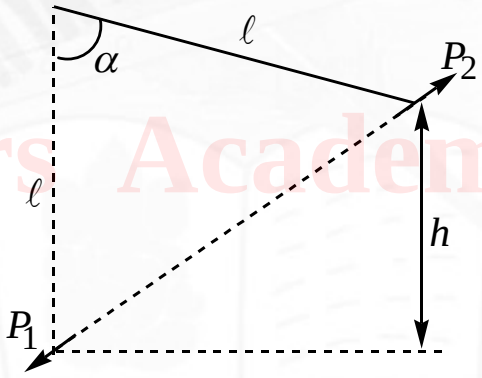
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
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Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

21. A small dipole of dipole moment $\vec{P}_2$ of mass m is suspended vertically by a string of length ℓ. A small dipole of dipole moment $\vec{P}_1$ is now brought towards $\vec{P}_2$ from infinity so that the P_2 moves away and gravitational potential energy of dipole P_1 does not change. The final equilibrium position of the system including the direction of dipole, the angles and distance is shown in figure (where g is the acceleration due to gravity). Choose the correct statement(s).



A) The work done in bringing the dipole to this position is $\frac{4mgh}{3}$.

B) The work done in bringing the dipole to this position is $\frac{5mgh}{3}$.

C) Change in electrostatic potential energy of dipole is $\frac{mgh}{3}$.

D) Change in electrostatic potential energy of dipole is $\frac{2mgh}{3}$.

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22. An X-ray tube operating at 84 kV & 10 mA. Only 1% of the electric power supplied is converted into X-rays. If tangent has a mass of 300 gm & specific heat of 0.04 cal/gm/$^{\circ}$C. Then choose the correct statements (use 1 cal = 4.2 J & $hc = 12400\ eV - \mathring{A}$) Assume entire thermal energy is observed by the target.

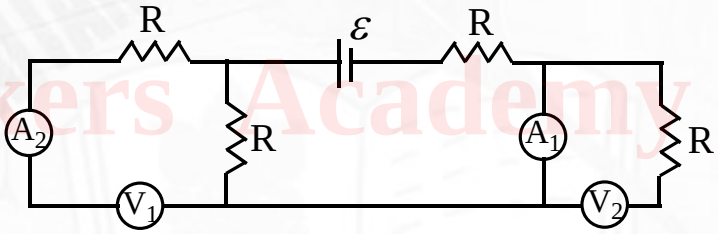
A) Cut-off wavelength is approximately $0.15\mathring{A}$.

B) Cut-off wavelength is approximately $6.8\mathring{A}$.

C) Average rate of temperature rise of target (assuming no heat loss) is $16.5^{o}C/\sec$.

D) Average rate of temperature rise of target (assuming no heat loss) is $1.7^{o}C/\sec$.

23. Consider the following electric circuit that comprises of a battery of emf E = 30V and negligible internal resistance and identical ammeters and voltmeters. All the resistors have $R = 50\Omega$. Select the correct alternative (s). (consider ammeters and voltmeters are ideal).



A) The readings are $A_1 = 0.3A; A_2 = 0; V_1 = 0; V_2 = 15V$.

B) If A_1 and V_1 are interchanged and so are A_2 and V_2, then the readings become $A_1 = 0.2A; A_2 = 0; V_1 = 10V; V_2 = 10V$.

C) If A_1 and V_2 are interchanged then the readings become $A_1 = 0A; A_2 = 0; V_1 = 30V; V_2 = 30V$.

D) If A_1 and V_2 are interchanged then the readings become $A_1 = 0A; A_2 = 0; V_1 = 30V; V_2 = 10V$.

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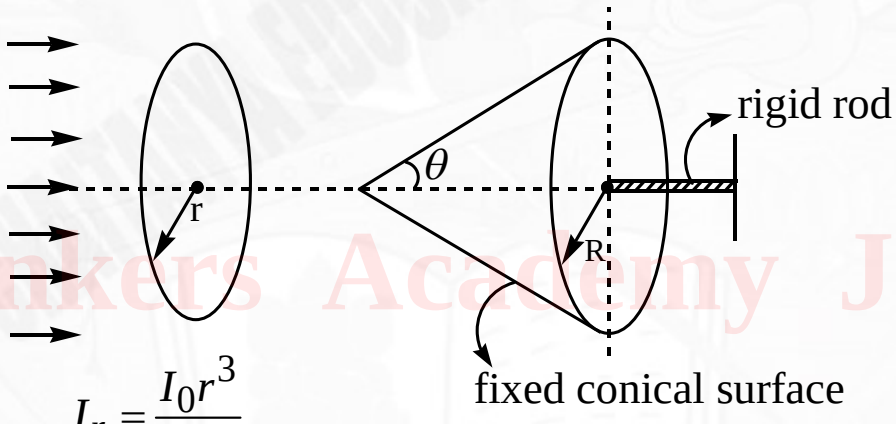
SECTION-III
(NUMERICAL VALUE)

This section contains **SIX (06)** questions.

- The answer to each question is a NUMERICAL VALUE .
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases..

24. A cylindrical beam of light having a variable intensity I_r is incident on the curved surface of a fixed cone of base radius R. Intensity of beam changes with radial distance r, according to the function $I_r = \frac{I_0 r^3}{R^3}$. Conical surface is completely reflecting surface. If the force exerted by the light on the cone is of the form $\frac{k_1 \pi I_0 R^2 \sin^2\theta}{k_2 C}$ in simplest ratio, where k_1 and k_2 are integers and C is the speed of light, then value of $(k_1 + k_2)$ will be?



$$I_r = \frac{I_0 r^3}{R^3}$$

25. When a resistance R is connected in series with an element A, the electric current is found to be lagging behind the applied AC voltage V by an angle 30°. When the same resistance is connected in series with element B, current leads by 60°. When R, A, B are connected in series, the current now leads voltage by θ which is equal to $\tan^{-1}\left(K/\sqrt{3}\right)$, then the value of K is (assume same AC source is used in all cases)

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26. The angular width $\Delta\theta$ of the central maximum in the diffraction pattern of a slit illuminated by light of wavelength 500 nm is measured. The angular width of the central maximum is found to increase by 25% when the same slit is illuminated by another light of wavelength λ' (in nm) is_____

27. A proton with rest mass of $938272.0 keV/c^2$ and kinetic energy of 1MeV hits a nucleus of the isotope ${}_3Li^7$ with mass of $7m_a$ and induces a decay to two non – excited α particles with rest mass of $3.727379 GeV/c^2$. What will be total combined kinetic energy in MeV of these two particles? Consider $m_a = 931.2720 MeV/c^2$ and $c = 299792458 m.s^{-1}$.

28. In order to impart an angular velocity to an earth satellite, the geomagnetic field can be used. Find the maximum possible angular velocity (in $10^{-2} rad/s$) about its own axis gained by the satellite if a storage battery with a capacity of $Q = 5 Amp - hours$ is discharged suddenly through a coil $N = 20$ turns wound around the satellite's surface along the circumference of the largest circle. The satellite has a mass of $m = 10^3 kg$ and is a thin walled uniform sphere. The geomagnetic field is parallel to the winding plane and its flux density is $B = 0.5 Gauss$. (Take $\pi = 3.14$)

29. Let $\lambda_1 \& \lambda_2$ be the De-Broglie wavelength of the two bodies of mass M and 2M revolving about their common centre of mass under the action of mutual gravitational force of attraction respectively. The value of $\frac{\lambda_1}{\lambda_2} = k$, find the value of k.



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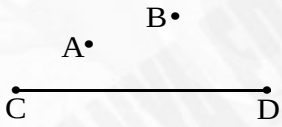
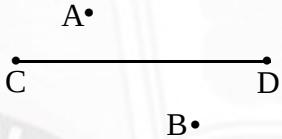
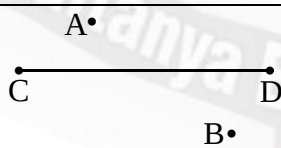
SECTION – IV
(MATCHING TYPE)

This section contains **THERE (03)** Matching List Sets.

- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases.

30. In Column-I the position of real object (A or B) and the position of its (real or virtual) image (B or A) and the optical axis (CD) are given. In Column-II the optical elements that can form the images given in Column-I are given. (In all the cases the distance of B is larger than that of A from the line CD) Match the Column-I with Column-II and mark the best possible option.

	Column-I		Column-II
a)	 CD is the optical axis of lens	p)	Convex lens
b)	 CD is the optical axis of mirror	q)	Concave lens
c)	 CD is the optical axis of lens	r)	Convex mirror
d)	 CD is the optical axis of mirror	s)	Concave mirror

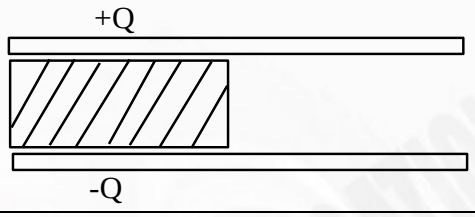
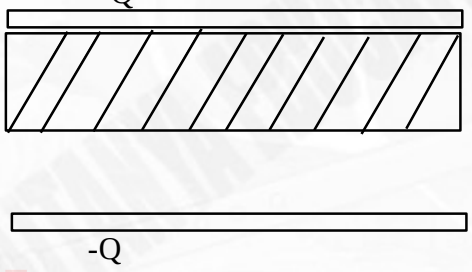
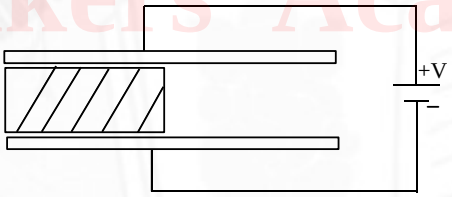
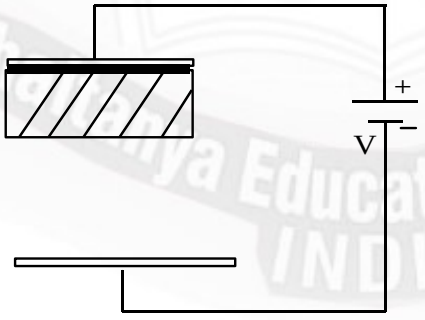
A) $a-p,q;b-r,s;c-p:d-s$ **B)** $a-p;b-r;c-q:d-s$

C) $a-q;b-s;c-p:d-r$ **D)** $a-p;b-s;c-p:d-r$

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31. In column I a parallel plate capacitor with half of its volume filled with a dielectric material of dielectric constant 4 in different configurations are shown. In first two cases (a and b) the upper and lower plates of capacitor are given charges +Q and –Q respectively and in the last two cases (c and d) the capacitor plates are connected to terminals of a battery. In all the cases E1 is the electric field inside the dielectric material and E_2 is the electric field in the remaining half space between the capacitor plates. In Column-II, the ratio of E_1 / E_2 is given for different situations of Column-I

Match the Column-II with correct options of Column-I

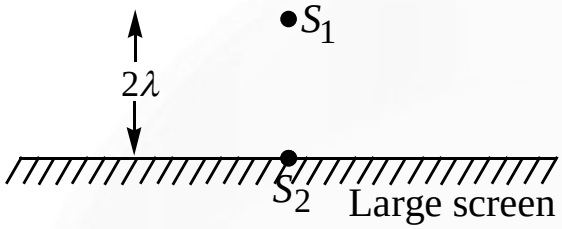
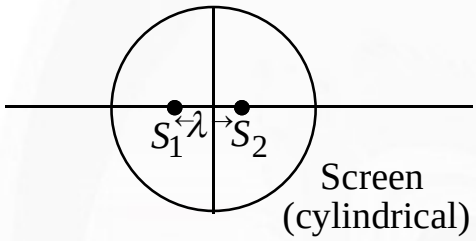
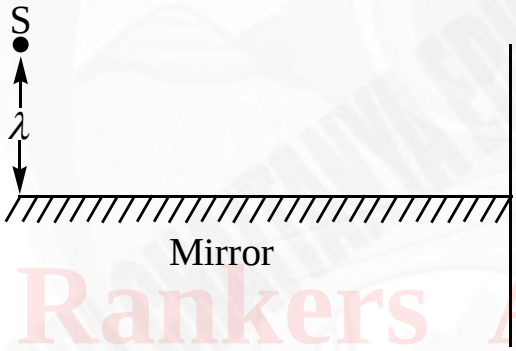
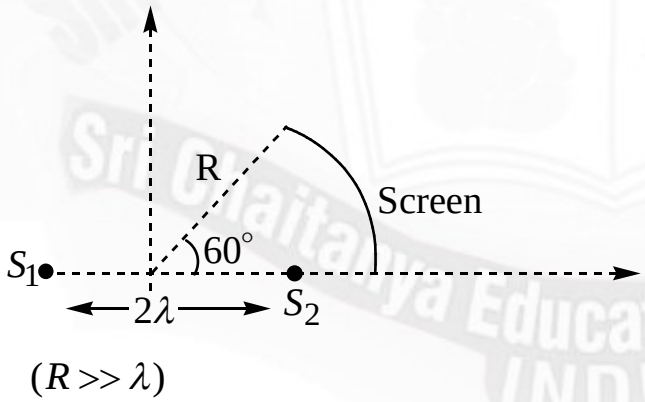
	Column-I		**Column-II**
a)		p)	1
b)		q)	4
c)		r)	0.5
d)		s)	0.25

A) $a-p;b-q;c-r;d-s$ **B)** $a-q;b-s;c-p;d-r$

C) $a-s;b-p;c-s;d-p$ **D)** $a-p;b-s;c-p;d-s$

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32. S_1 and S_2 in list I represent coherent points sources, S represents a point source. $\lambda =$ wavelength of light emitted by the sources. Fringe pattern is observed on the screen. Exclude the position of source while detecting the fringe pattern.

List – I		List – II	
A)		P)	Number of maxima = 2 and number of minima=1
B)		Q)	Number of maxima = 2 and number of minima= 4
C)		R)	Number of maxima = 4 and number of minima = 4
D)		S)	Number of maxima = 2 and number of minima = 2

A) $A-P, B-R, C-S, D-Q$ **B)** $A-R, B-Q, C-S, D-P$

C) $A-Q, B-S, C-R, D-P$ **D)** $A-Q, B-R, C-S, D-P$

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CHEMISTRY **Max Marks: 60**

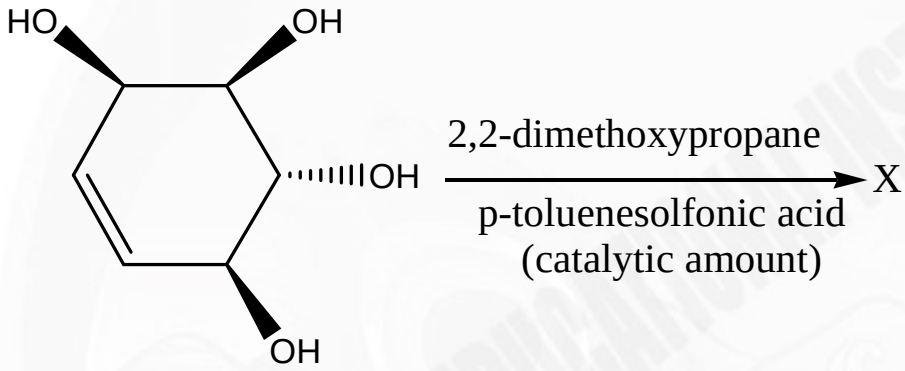
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

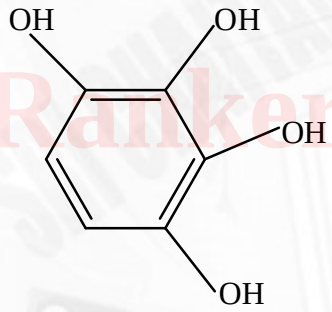
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

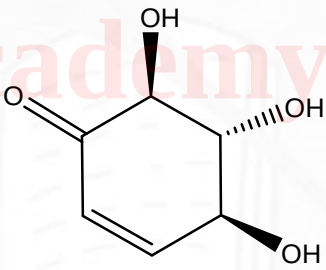
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

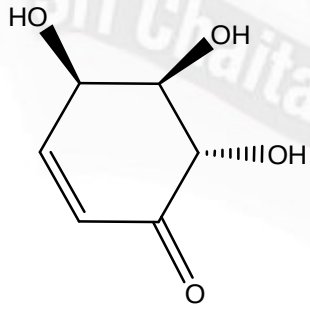
33. The major product X formed in the reaction given below is

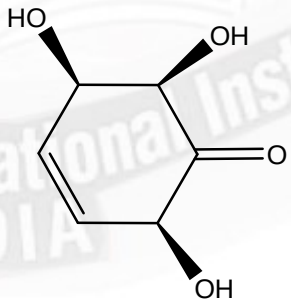


Oxidation of the product X, obtained in the above reaction, with active manganese dioxide, followed by acidic hydrolysis gives.

A)


B)


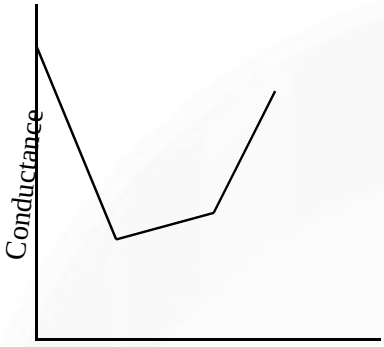
C)


D)


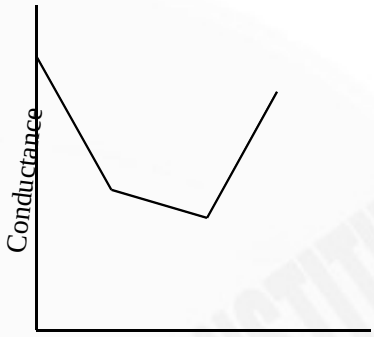
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34. Which of the following correctly represents the conductometric titration of an equimolar mixture of acetic acid and HCl against NaOH?

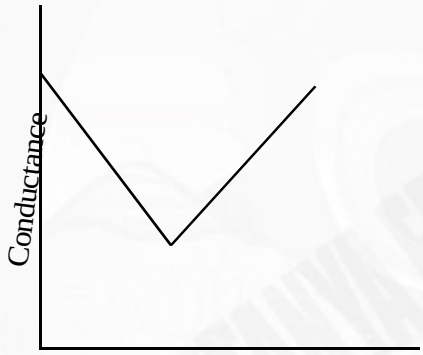
A)



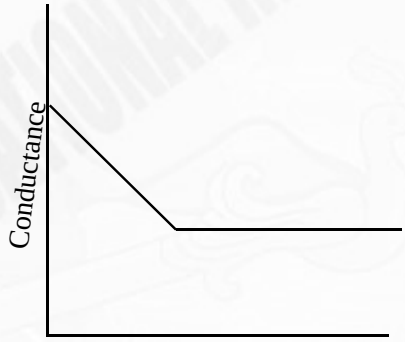
B)



C)



D)



35. Which of the following statements is incorrect?

A) Sodium dichromate is more soluble than potassium dichromate.

B) V_2O_5 reacts with alkalis as well as acids to give VO_4^{3-} and VO_2^{+} respectively.

C) IE_1 of $V < Ti$, but IE_2 of $V > Ti$ (IE – Ionization enthalpy)

D) IE_1 of $Cr < Mn$, but IE_2 of $Cr > Mn$ and IE_3 of $Cr > Mn$ (IE – Ionization enthalpy)

36. Which of the following statements or reactions is incorrect?

A) Among PCl_5, AsH_4^{+}, XeF_4 and XeO_2F_2; in XeF_4 maximum atoms are lie in the same plane.

B) The element with $[Xe]\,4f^{14}\,5d^{1}\,6s^{2}$ belongs to 6th period and group 3 of the modern periodic table

C) The number of stereo isomers possible for $\left[Co(NH_3)\,Cl\,(en)_2\right]^{2+}$ is 3

D) $Au \xrightarrow{Aqua\ regia} \left[AuCl_2\right]^{-}$ + other products

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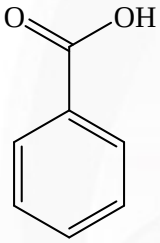
SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

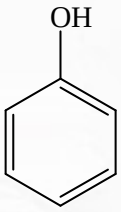
This section contains **THERE (03)** questions.

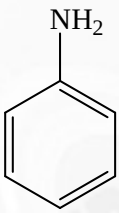
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

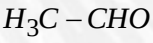
Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

37.


(P)


(Q)


(R)


(S)

Considering above compounds, identify the correct statement(s)?

A) Compound (P) and Compound (Q) can be differentiated by $NaHCO_3$ as well as aq. NaOH.

B) Compound (Q) and Compound (R) can be differentiability by Neutral $FeCl_3$.

C) Compound (R) and Compound (S) can be differentiated by $NaHSO_3$.

D) Compound (R) upon reaction with $AcCl$ / $Pyridine$ gives benzanilide as the major product

38. Which of the following statements are correct:-

A) The unit cell having crystal parameters $a = b = c, and\ \alpha = \beta = \gamma \neq 90°$ is trigonal

B) In case of metal excess defect density of the crystal always increases

C) In case of impurity defect density of the crystal may decrease or Increase

D) ZnS type crystal normally shows frankel defect

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39. In which of the following options, complex(es) is/are correctly matched with given information?

A) $\left[Cu\left(NH_3\right)_4\right]^{2+}$, *tetrahedral, paramagnetic*

B) $\left[Ni\left(CO\right)_4\right]$, *square planar, diamagnetic*

C) $\left[Co\left(CN\right)_6\right]^{4-}$, *number of unpaired electrons* $=1$

D) $\left|CFSE\right|$ *of* $\left[Co\left(C_2O_4\right)_3\right]^{3-} = 2\times\left|CFSE\right|$ *of* $\left[Cr\left(C_2O_4\right)_3\right]^{3-}$

(CFSE= Crystal field stabilization energy) (Ignore pairing energy)

SECTION-III

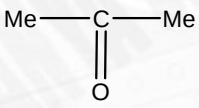
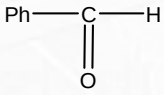
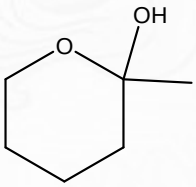
(NUMERICAL VALUE)

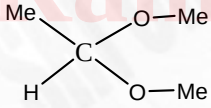
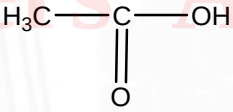
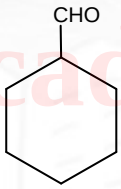
This section contains **SIX (06)** questions.

- The answer to each question is a NUMERICAL VALUE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme

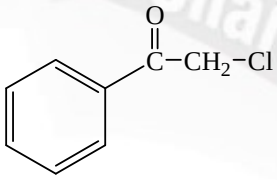
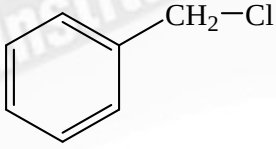
Full Marks : +4 If ONLY the correct integer is entered; Zero Marks : 0 In all other cases.

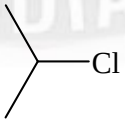
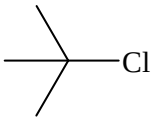
40. How many of the following will give positive Fehling test:

a) HCHO b)  c)  d) 

e)  f)  g)  h) HCOOH

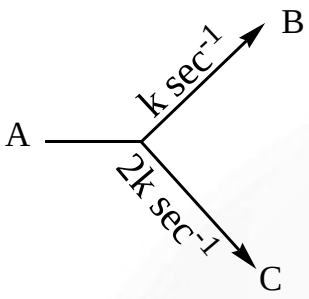
41. How many compounds are more reactive towards S_N1 as well as S_N2 than $CH_3-CH_2-CH_2-Cl$:

i)  **ii)** CH_3-O-CH_2-Cl **iii)** 

iv) $CH_3-CH_2-\ddot{N}H-CH_2-Cl$ **v)**  **vi)** $CH_2=CH-CH_2-Cl$ **vii)** 

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42. In a parallel first order reaction starting with 1 mole 'A' as



If A, B & C are optically active with standard optical rotation $15^0, 10^0$ & -20^0 degree per mole respectively then give time in sec when solution will be optically inactive.

Given: $k = 0.024 \sec^{-1}$, $\ln 2 = 0.7, \ln 5 = 1.60$.

43. For the mechanism

$A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} C + D$ (Step-I)

$2C \xrightarrow{k_2} G + H$ (Step-II)

Step-II is rate determining step. Given the activation energies $E_{a_1} = 120 \text{kJ / mole}$, $E_{a_{-1}} = 96 kJ / mole$ and $Ea_2 = 196 kJ /$ mole. Find out E_a for the overall reaction in kJ. $[2A + 2B \rightarrow 2D + G + H]$.

44. Consider the following observations and find the value of $(a + b + c)$?

Observations I:

a = Total number of black/brown coloured compounds among PbS, CuS, HgS, Ag_2S, NiS, MnS, CoS, $MnO(OH)_2$, $HgO.HgNH_2I$ and $Mn(OH)_2$.

Observations II:

Oxidation states of Mn in the products obtained when manganate ion is dispropationate in a neutral or acidic solution are "+b" and "+c"

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45. Consider the following reactions:

I) Native ore of Ag $\xrightarrow[H_2O,O_2]{CN^-(aq)}$ complex 'X' + other products

complex 'X' $\xrightarrow{Zn}$ complex 'Y' + other products.

II) Native ore of Au $\xrightarrow[H_2O,O_2]{CN^-(aq)}$ complex 'P' + other products

complex 'P' $\xrightarrow{Zn}$ complex 'Q' + other products.

Then the value of $(a+b+c)$ is_____.

Given that:

a = Number of unpaired electrons in 'X'.

b = Number of unpaired electrons in 'P'.

c = Number of electrons present in t_2 set of orbitals of complex 'Q'.

SECTION – IV
(MATCHING TYPE)

This section contains **THERE (03)** Matching List Sets
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists : **List-I** and **List-II.**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and **List-II** has **Five** entries (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks:+3 **ONLY** if the option corresponding to the correct combination is chosen;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: –1 In all other cases..

46. Match each species in List-I with List-II and choose correct option

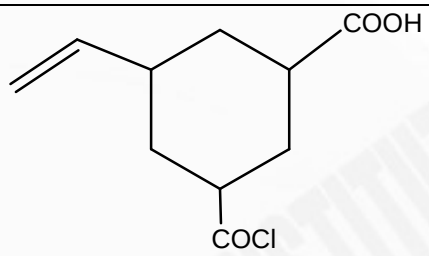
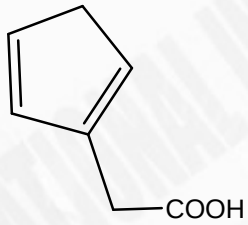
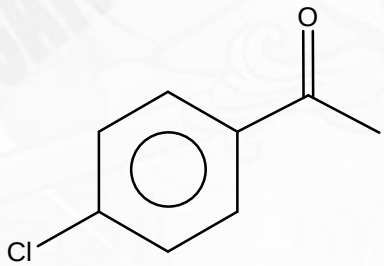
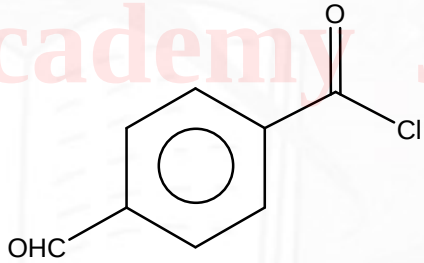
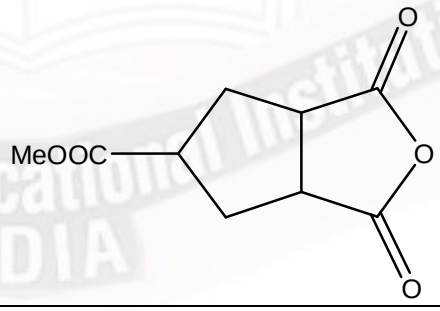
List-I		List-II	
(I)	A carbohydrate which yields only glucose on hydrolysis	(P)	Sucrose
(II)	A carbohydrate which yields glucose and fructose on hydrolysis	(Q)	Lactose
(III)	A carbohydrate which yields glucose and galactose on hydrolysis	(R)	Amylose
(IV)	A carbohydrate which can reduce Fehling's solution	(S)	Mannose
		(T)	Cellulose

A) $I \to R,T; II \to P; III \to Q; IV \to Q,S,R$ **B)** $I \to RT; II \to Q; III \to P; IV \to QS$

C) $I \to R; II \to P; III \to Q; IV \to QST$ **D)** $I \to R,T; II \to P; III \to Q; IV \to Q,S$

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47. Match List-I with List-II.

	List-I		List-II
	(Number of equivalents of *MeMgBr* consumed when excess is used)		(Compounds)
P)	2	1)	
Q)	3	2)	
R)	1	3)	
S)	4	4)	
		5)	

A) $P \to 2; Q \to 1,4; R \to 3; S \to 5$ **B)** $P \to 2,4; Q \to 3; R \to 1; S \to 5$

C) $P \to 1,4; Q \to 3; R \to 2; S \to 5$ **D)** $P \to 5; Q \to 1; R \to 2; S \to 3$

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48. Matrix matching

Column-I (Solution)		Column-II (Property)	
A)	A solution of vapour pressure 110 mm, containing one mole of liquid A $(P^0 = 150mm)$ and two moles of liquid B $(P^0 = 75mm)$	P)	Solvent-solvent, solute-Solute attractive forces are stronger than solvent-solute attractive force
B)	A solution of vapour pressure 100 mm containing a liquid 'X' $(P^0 = 120mm)$ & liquid 'Y' $(P^0 = 80mm)$ with 60 mole percent of 'X' in vapour above the solution	Q)	ΔH of mixing is +ve
C)	A solution of vapour pressure 100 mm, containing 3 moles of liquid M $(P^0 = 150mm)$ and 2 moles of liquid 'N' $(P^0 = 150mm)$	R)	Solution obeys Raoults law
D)	A solution of vapour pressure 110 mm containing 2 moles of liquid 'E' $(P^0 = 90mm)$ and 4 moles of liquid F $(P^0 = 120mm)$	S)	Solvent-Solvent, Solute-Solute attractive forces are weaker than solvent–solute attractive force
		T)	ΔH of mixing is negative

A) $A-PQ, B-ST, C-R, D-R$ **B)** $A-PQ, B-R, C-ST, D-R$

C) $A-ST, B-R, C-PQ, D-R$ **D)** $A-R, B-R, C-PQ, D-ST$

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